National University of Computer and Emerging Sciences

Peshawar Campus

Data Structures (CS2001)

Sessional-II Exam

Date: 4th November 2024

Course Instructor(s)

Waqas Ali / Qasim Jan

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 3

Roll No _____ Section _____ Student Signature _____

Do not write below this line

Attempt all the questions.

CLO 1: Understand basic concepts of data structure and algorithms.

Q1: Write a function for a stack of integers that reverses either the even or odd values, while keeping all other elements in their original positions. The function should take a boolean parameter named **even** to specify which values to reverse: If even is true, reverse only the even integers in the stack. If even is false, reverse only the odd integers. **[10]**

Consider the following example:

Consider the stack with top at 8:

8	1	2	5	4	3	6	7

The stack after evens reversed (top at 6):

6	1	4	5	2	3	8	7

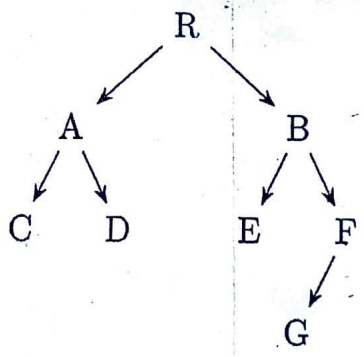
The stack after odds reversed (top at 8):

8	7	2	3	4	5	6	1

Note: You are allowed to use any additional data structure.

CLO 3: Design appropriate data structures to solve real world problems related to the program.

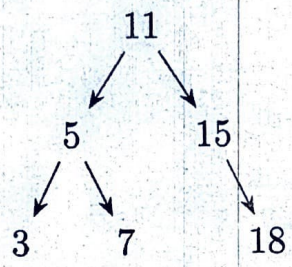
Q2: Write a recursive function, that finds and returns the number of nodes in the longest path in a binary tree. Consider the following binary tree, the longest path in this tree is **R -> B -> F -> G** which contains 4 nodes therefore, the function should return 4. **Note:** The use of a global or static count variable is not allowed. **[10]**



CLO 3: Design appropriate data structures to solve real world problems related to the program.

Q3: Write a recursive function that finds and prints the range of values in a binary search tree that fall within a given range [low, high]. The **low** and **high** values of the range should be passed to the function as parameters. **[10]**

Consider the following binary search tree:



For the range [7, 15], the output should be: [7, 11, 15]

For the range [6, 11], the output should be: [7, 11]

---------- The End ----------